Module 5: State Estimation in Multicopter Control

Dr. Juan Augusto Paredes Salazar, Dr. Ankit Goel
Mechanical Engineering Department, UMBC
{japarede, ankgoel}@umbc.edu

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1 State Estimation for Linear Systems

1.1 Motivation

In many practical systems, the full state is not directly measurable. Instead, sensors provide partial information about the state through the output

$$y = Cx. \tag{1}$$

Thus, the problem of state estimation is:

Given the input $u(t)$ and measured output $y(t)$, reconstruct the full state $x(t)$.

This arises in many systems:

- A pendulum where only the angle θ is measured but $\dot{\theta}$ is not,
- Mechanical systems where velocity is not directly sensed,
- UAVs where only partial states (e.g., position, yaw, angular velocity) are available.

1.2 Interpretation of Measurements

The output equation

$$y = Cx \tag{2}$$

represents a linear combination of the state variables.

Each row of C extracts or combines specific components of the state. For example:

- $C = [1\ 0\ \cdots\ 0]$ measures the first state directly,
- $C = [0\ 1\ \cdots\ 0]$ measures the second state,
- more general C corresponds to combinations of states.

Thus, the measurements contain *information about the state*, but may not uniquely determine it unless the system is observable.

1.3 System Model

Consider the linear system

$$\dot{x} = Ax + Bu, \tag{3}$$
$$y = Cx, \tag{4}$$

where $x \in \mathbb{R}^n$ is the state and $y \in \mathbb{R}^p$ is the measured output.

1.4 Naive State Estimator

A natural first attempt is to simulate the system dynamics:

$$\dot{\hat{x}} = A\hat{x} + Bu. \tag{5}$$

Define the estimation error

$$e = x - \hat{x}. \tag{6}$$

Then,

$$\dot{e} = Ae. \tag{7}$$

Key observation:

- If A is stable, then $e(t) \to 0$.
- If A is unstable, the estimation error diverges.

Thus, a pure model-based estimator is insufficient in general.

1.5 Observer Design (Luenberger Observer)

To correct the prediction, we inject measurement feedback into the estimator:

$$\dot{\hat{x}} = A\hat{x} + Bu + F(y - C\hat{x}), \tag{8}$$

where $F \in \mathbb{R}^{n\times p}$ is the observer gain.

The term

$$y - C\hat{x} \tag{9}$$

is called the **innovation** and represents the mismatch between measured and predicted outputs.

1.6 Error Dynamics

Define the estimation error

$$e = x - \hat{x}. \tag{10}$$

Then,

$$\begin{aligned} \dot{e} &= \dot{x} - \dot{\hat{x}} && (11)\\ &= (Ax + Bu) - (A\hat{x} + Bu + F(y - C\hat{x})) && (12)\\ &= (A - FC)e. && (13) \end{aligned}$$

Interpretation:

- A propagates the error,
- $F(y - C\hat{x})$ corrects the error using measurements,
- F acts as a feedback gain for the estimation error.

1.7 Observer Stability

The estimation error evolves as

$$\dot{e} = (A - FC)e. \tag{14}$$

Proposition 1.1. *The estimation error converges to zero if and only if $(A - FC)$ is Hurwitz.*

Thus, observer design reduces to placing the eigenvalues of $(A - FC)$.
Observer design converts the state estimation problem into a stabilization problem.

- The estimation error satisfies
$$\dot{e} = (A - FC)e,$$
which is independent of the input u.
- Thus, designing an observer reduces to stabilizing this error system.

Key insight:

State estimation is achieved by stabilizing the error dynamics using output feedback.

1.8 Effect of Initial Conditions

The observer does not require knowledge of the true initial state. Even if

$$\hat{x}(0) \neq x(0),$$

the estimation error evolves as

$$\dot{e} = (A - FC)e,$$

and converges to zero if $(A-FC)$ is Hurwitz. Thus, the observer asymptotically corrects any initial mismatch.

1.9 Observability

Observer design is possible only if the system is observable.

Definition 1.1 (Observability)**.** The pair (A, C) is observable if the initial state $x(0)$ can be uniquely determined from the output $y(t)$ over a finite time interval.

The observability matrix is

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}. \tag{15}$$

Proposition 1.2. *The pair (A, C) is observable if and only if*

$$\text{rank}(\mathcal{O}) = n. \tag{16}$$

1.10 Observer Design via Duality

Observer design is dual to state-feedback control.

We use the identity

$$\operatorname{eig}(A - FC) = \operatorname{eig}(A^T - C^T F^T). \tag{17}$$

Thus, observer gains can be computed using pole placement:

$$F = \left[\texttt{place}(A^T, C^T, p_{\text{obs}})\right]^T. \tag{18}$$

This requires (A, C) to be observable.

Proposition 1.3. *The pair (A, C) is observable if and only if the pair (A^T, C^T) is controllable.*

Thus, observer design via pole placement is possible if and only if (A, C) is observable.

1.11 Computing the Observer Gain F Using `place` and `lqr`

In practice, the observer gain F is computed numerically using standard control design tools. Two common approaches are pole placement and LQR-based design.

1.11.1 Using Pole Placement (`place`)

Using the duality between observability and controllability, the observer gain can be computed as

$$F = \left[\texttt{place}(A^T, C^T, p_{\text{obs}})\right]^T, \tag{19}$$

where p_{obs} is the desired set of observer poles.

Procedure:

- Choose desired observer poles p_{obs}.
- Typically, these are placed faster than the controller poles.
- Compute
$$F = \left[\texttt{place}(A^T, C^T, p_{\text{obs}})\right]^T.$$

Remark:

- Faster poles $\Rightarrow$ faster convergence of estimation error.
- Very fast poles $\Rightarrow$ amplification of measurement noise.

1.11.2 Using LQR (`lqr`) for Observer Design

An alternative is to compute F using an LQR formulation applied to the dual system. Consider the dual system

$$\dot{z} = A^T z + C^T u. \tag{20}$$

We define a quadratic cost

$$J = \int_0^\infty \left(z^T Q z + u^T R u\right) dt. \tag{21}$$

Then the optimal gain is

$$K_{\text{lqr}} = \texttt{lqr}(A^T, C^T, Q, R), \tag{22}$$

and the observer gain is

$$F = K_{\text{lqr}}^T. \tag{23}$$

Interpretation:

- Q penalizes estimation error,
- R penalizes sensitivity to measurement noise,
- The ratio Q/R determines the speed of convergence.

Tuning Guidelines:

- Larger $Q \Rightarrow$ faster observer (more aggressive correction),
- Larger $R \Rightarrow$ slower observer (less noise amplification).

1.11.3 Comparison

- `place`: direct pole assignment, intuitive but requires manual tuning.
- `lqr`: systematic, trades off convergence speed and noise sensitivity.

Key Insight:

Observer design can be performed using the same tools as state-feedback control, applied to the dual system.

1.12 Observer Implementation

The observer can be rewritten as

$$\dot{\hat{x}} = (A - FC)\hat{x} + Bu + Fy. \tag{24}$$

Interpretation:

- $(A - FC)\hat{x}$: internal dynamics,
- Bu: known input,
- Fy: correction using measurements.

Note that faster observer poles lead to faster convergence but increase sensitivity to measurement noise.

1.13 Connection to Control

Full-state feedback assumes access to x,

$$u = Kx. \tag{25}$$

If the full state is not available, then we can use

$$u = K\hat{x}. \tag{26}$$

1.14 Observer-Based Control

Combining the system and observer:

$$\dot{x} = Ax + Bu, \tag{27}$$
$$\dot{\hat{x}} = A\hat{x} + Bu + F(y - C\hat{x}), \tag{28}$$
$$u = K\hat{x}. \tag{29}$$

1.15 Separation Principle

Consider the plant

$$\dot{x} = Ax + Bu, \tag{30}$$
$$y = Cx, \tag{31}$$

together with the observer-based controller

$$u = K\hat{x}, \tag{32}$$
$$\dot{\hat{x}} = A\hat{x} + Bu + F(y - C\hat{x}). \tag{33}$$

The estimation error satisfies

$$\dot{e} = (A - FC)e. \tag{34}$$

Also, since $\hat{x} = x - e$, the control law becomes

$$u = K\hat{x} = K(x - e). \tag{35}$$

Substituting into the plant dynamics gives

$$\dot{x} = Ax + BK(x - e) \tag{36}$$
$$= (A + BK)x - BKe. \tag{37}$$

Therefore, the combined dynamics in the coordinates (x, e) are

$$\frac{d}{dt}\begin{bmatrix} x \\ e \end{bmatrix} = \underbrace{\begin{bmatrix} A + BK & -BK \\ 0 & A - FC \end{bmatrix}}_{A_{\text{aug}}} \begin{bmatrix} x \\ e \end{bmatrix}. \tag{38}$$

Since A_{aug} is block upper triangular, its eigenvalues are exactly the union of the eigenvalues of its diagonal blocks. Hence,

$$\text{eig}(A_{\text{aug}}) = \text{eig}(A + BK) \cup \text{eig}(A - FC). \tag{39}$$

Thus, the controller poles and the observer poles appear independently in the augmented closed-loop dynamics.

Key implication:

- The state-feedback gain K determines the eigenvalues of $A + BK$,
- The observer gain F determines the eigenvalues of $A - FC$,
- The controller and observer can therefore be designed independently.

1.16 Design Guidelines

- Choose observer poles faster than controller poles.
- Avoid excessively fast poles (noise amplification).
- Ensure (A, C) is observable before design.

1.17 Key Insight

State estimation reconstructs unmeasured states by combining a predictive model with measurement correction. The observer acts as a dynamic filter that continuously corrects model predictions using sensor data.

2 State Estimation for a Bicopter

2.1 Nonlinear Bicopter Model

Consider the planar bicopter shown in Figure 1, with dynamics

$$\ddot{p}_1 = -\frac{\sin\theta}{m}T, \tag{40}$$

$$\ddot{p}_3 = -\frac{\cos\theta}{m}T + g, \tag{41}$$

$$\ddot{\theta} = \frac{1}{J}M, \tag{42}$$

where

- p_1, p_3 are the horizontal and vertical positions,
- θ is the tilt angle,
- T is the total thrust,
- M is the total moment.

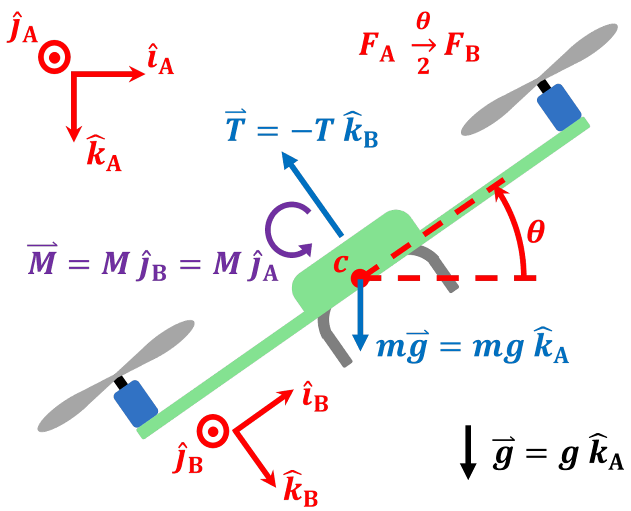

Figure 1: Diagram of a bicopter.

2.2 Linearization About Hover

We linearize about the hover equilibrium

$$p_1 = p_{1,\mathrm{eq}}, \quad p_3 = p_{3,\mathrm{eq}}, \quad \dot{p}_1 = \dot{p}_3 = 0, \quad \theta = 0, \quad \dot{\theta} = 0, \quad T = mg, \quad M = 0,$$

where $p_{1,\mathrm{eq}}, p_{3,\mathrm{eq}}$ represent an arbitrary position.

The linearized dynamics are

$$\ddot{\tilde{p}}_1 = -g\,\tilde{\theta}, \tag{43}$$

$$\ddot{\tilde{p}}_3 = -\frac{1}{m}\tilde{T}, \tag{44}$$

$$\ddot{\tilde{\theta}} = \frac{1}{J}\tilde{M}. \tag{45}$$

Define the state and input

$$x = \begin{bmatrix} \tilde{p}_1 \\ \tilde{p}_3 \\ \dot{\tilde{p}}_1 \\ \dot{\tilde{p}}_3 \\ \tilde{\theta} \\ \dot{\tilde{\theta}} \end{bmatrix}, \quad u = \begin{bmatrix} \tilde{T} \\ \tilde{M} \end{bmatrix}.$$

Then

$$\dot{x} = Ax + Bu,$$

with

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -g & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{J} \end{bmatrix}.$$

2.3 Sensor Configurations and Observability

We now examine how different measurement choices affect observability by computing the observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}.$$

2.3.1 Case 1: Measure only θ

$$y = \theta, \quad C = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Then

$$CA = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad CA^2 = 0.$$

Thus,

$$\mathcal{O} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$\mathrm{rank}(\mathcal{O}) = 2 < 6.$$

Conclusion: Not observable.

—

2.3.2 Case 2: Measure only p_1

$$y = p_1, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Compute:

$$CA = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix},$$

$$CA^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & -g & 0 \end{bmatrix},$$
$$CA^3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -g \end{bmatrix}.$$

Thus,

$$\mathcal{O} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -g & 0 \\ 0 & 0 & 0 & 0 & 0 & -g \end{bmatrix}.$$

$$\text{rank}(\mathcal{O}) = 4 < 6.$$

Conclusion: Not observable.
—

2.3.3 Case 3: Measure p_1 and θ

$$y = \begin{bmatrix} p_1 \\ \theta \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Compute key rows:

$$CA = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$
$$CA^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & -g & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Collecting independent rows, we obtain full rank:

$$\text{rank}(\mathcal{O}) = 4 < 6.$$

Conclusion: Not observable.
—

2.3.4 Case 4: Measure p_1 and p_3

$$y = \begin{bmatrix} p_1 \\ p_3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Compute:

$$CA = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix},$$
$$CA^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & -g & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$
$$CA^3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -g \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The resulting observability matrix has rank

$$\text{rank}(\mathcal{O}) = 6.$$

Conclusion: Observable.
—

2.3.5 Case 6: Measure p_1, p_3, and $\dot{\theta}$

$$y = \begin{bmatrix} p_1 \\ p_3 \\ \dot{\theta} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Computing the observability matrix, we obtain independent rows corresponding to

$$p_1, \quad p_3, \quad \dot{p}_1, \quad \dot{p}_3, \quad \theta, \quad \dot{\theta}.$$

Thus,

$$\text{rank}(\mathcal{O}) = 6.$$

Conclusion: The system is observable.

Remark: This sensor configuration is physically realistic: position measurements can be obtained from vision or GPS, while $\dot{\theta}$ is directly measured using a gyroscope. The combination provides sufficient information to reconstruct all states.

2.3.6 Summary

- Observability depends on both the measurement selection and the system dynamics.
- A single measurement is insufficient for full-state observability.
- Not all two-sensor configurations are sufficient. For example:
  - (p_1, θ) does *not* yield an observable system,
  - (p_1, p_3) yields an observable system.
- Measuring position and angular rate,

  $$y = \begin{bmatrix} p_1 \\ p_3 \\ \dot{\theta} \end{bmatrix},$$

  provides a physically realistic and fully observable configuration.
- Observability is enabled by dynamic coupling between states; measurements need not include every state directly, but must collectively excite all modes of the system.

2.4 Key Insight

Even when not all states are measured, the system dynamics allow reconstruction of hidden states. Observability depends not only on sensors, but also on how states are coupled through the dynamics.

2.5 Observer pole placement for control.

When the bicopter observer is used together with a feedback controller, the observer convergence rate should be chosen relative to the controller bandwidth. In a typical cascaded bicopter or multicopter architecture, the rotational dynamics form the fast inner loop, while the translational dynamics form the slower outer loop. Therefore, the observer modes associated with the rotational states, especially θ and $\dot{\theta}$, should generally be placed faster than the inner-loop attitude-control poles. This ensures that the attitude controller acts on sufficiently accurate estimates of the rotational states.

At the same time, observer poles should not be placed arbitrarily far to the left. Excessively fast observer dynamics can amplify measurement noise, especially when angular-rate or position measurements are noisy. Thus, the observer should be faster than the relevant control loop, but not so fast that it becomes noise-sensitive or numerically fragile.

3 Specialization to Multicopters

3.1 Connection to Linear Observer Framework

The multicopter dynamics fit into the linear system framework

$$\dot{x} = Ax + Bu, \quad y = Cx, \tag{46}$$

with the state vector defined as

$$x \triangleq \xi = \begin{bmatrix} p \\ v \\ q \\ \omega \end{bmatrix}.$$

Thus, all observer design results from Section 1 apply directly.

3.2 Linearized Multicopter Model

Consider the linearized multicopter dynamics about hover:

$$\dot{\xi} = A\xi + Bv, \tag{47}$$

where the state is partitioned as

$$\xi = \begin{bmatrix} p \\ v \\ q \\ \omega \end{bmatrix}$$

A simplified linearized structure is

$$A = \begin{bmatrix} 0_3 & I_3 & 0_3 & 0_3 \\ 0_3 & 0_3 & G(\psi) & 0_3 \\ 0_3 & 0_3 & 0_3 & I_3 \\ 0_3 & 0_3 & 0_3 & 0_3 \end{bmatrix}, \quad B = \begin{bmatrix} 0_{3\times1} & 0_3 \\ H & 0_3 \\ 0_{3\times1} & 0_3 \\ 0_{3\times1} & J^{-1} \end{bmatrix},$$

where

$$G(\psi) = g \begin{bmatrix} -\sin(\psi) & -\cos(\psi) & 0 \\ \cos(\psi) & -\sin(\psi) & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{m} \end{bmatrix}.$$

3.3 Measurement Model

Typical onboard sensors provide:

- position p (GPS or vision),
- yaw ψ (magnetometer),
- angular velocity ω (gyroscope).

Thus,

$$y = \begin{bmatrix} p \\ \psi \\ \omega \end{bmatrix} = C\xi, \tag{48}$$

with

$$C = \begin{bmatrix} I_3 & 0 & 0 & 0 \\ 0 & 0 & e_3^{\mathrm{T}} & 0 \\ 0 & 0 & 0 & I_3 \end{bmatrix}.$$

3.4 Observability Analysis

To verify observability, consider the observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \end{bmatrix}.$$

We analyze how each state appears through successive multiplications:

- C directly measures:
$$p, \quad \psi, \quad \omega.$$
- CA introduces:
$$v \quad (\text{since } \dot{p} = v),$$
- CA^2 introduces:
$$(\phi, \theta) \quad \text{through translational acceleration coupling via } G,$$
- Higher powers do not introduce new independent directions.

Thus, all components of ξ appear in the rows of $\mathcal{O}$.

$$\operatorname{rank}(\mathcal{O}) = 12.$$

Conclusion: The system is observable under this sensor configuration.

3.5 Observer Design

The observer is constructed as

$$\dot{\hat{\xi}} = A\hat{\xi} + Bv + F(y - C\hat{\xi}). \tag{49}$$

3.6 Estimation Error Dynamics

Defining the estimation error

$$e = \xi - \hat{\xi}, \tag{50}$$

the error evolves as

$$\dot{e} = (A - FC)e. \tag{51}$$

Thus, the observer gain F determines the convergence rate of the estimation error.

3.7 Observer Design Guidelines

- Choose observer poles faster than controller poles,
- Avoid excessively fast poles (noise amplification),
- Ensure (A, C) is observable before design.

3.8 Physical Interpretation

- Velocity v is reconstructed from position measurements,
- Roll and pitch (ϕ, θ) are inferred from their effect on translational motion,
- Angular velocity ω provides direct rotational information.

3.9 Observer-Based Control

The control input is implemented as

$$u = K\hat{\xi}. \tag{52}$$

3.10 Limitations

- The observer is based on a linearized model and may degrade for large deviations from hover,
- Sensor noise enters through the innovation term $F(y - C\hat{\xi})$,
- Model mismatch can lead to estimation errors.

3.11 Key Insight

State estimation exploits dynamic coupling between translational and rotational dynamics. Even when some states are not directly measured, they can be reconstructed through their influence on measured variables.

4 Discrete-Time Kalman Filter

4.1 Historical Motivation

The Kalman filter is one of the most influential estimation algorithms in modern engineering. Its importance comes not only from its mathematical simplicity but also from its role in enabling major technological advances in which reliable state estimation was essential.

One of the earliest and most famous applications was the Apollo guidance and navigation program, in which Kalman filtering was used for spacecraft state estimation, sensor fusion, and trajectory correction during lunar missions [1]. Accurate estimation of position, velocity, and attitude from noisy measurements was critical for safe navigation and landing.

Kalman filtering also became fundamental in aerospace applications such as aircraft navigation, missile guidance, and satellite orbit determination, where sensors are noisy and direct measurement of the full state is impossible.

In modern engineering, the same ideas appear everywhere: GPS and inertial navigation systems, autonomous vehicles, robotics, drones, multicopters, financial forecasting, weather prediction, and industrial process control all rely on recursive state estimation methods derived from the Kalman filter. The central problem is always the same: the true system state cannot be measured perfectly, so we must combine imperfect models with noisy measurements to obtain the best possible estimate. The Kalman filter provides a systematic and optimal way to do exactly this.

4.2 Why Discrete-Time?

So far, we have formulated observers in continuous time

$$\dot{x} = Ax + Bu. \tag{53}$$

However, in practice,

- Sensors provide measurements at discrete time instants,
- Controllers are implemented on digital processors,
- Simulations (e.g., MATLAB, Simulink) operate in discrete time.

Thus, it is natural to work with a discrete-time model.

4.3 Kalman Filtering vs Classical Signal Filtering

In earlier courses, filtering is often introduced through signal-processing concepts such as low-pass filters (LPF), high-pass filters (HPF), and band-pass filters. These filters operate directly on measured signals to attenuate unwanted frequency content. In contrast, the Kalman filter is not a signal filter in the classical sense. Instead, it is a *model-based state estimator*. Key differences are summarized below.

Signal filtering vs state estimation. Classical filters operate on signals, that is,

$$y_{\text{filtered}}(t) = \mathcal{F}(y(t)),$$

where $\mathcal{F}$ is designed to remove noise in specific frequency bands. On the other hand, the Kalman filter operates on a *dynamical system model*

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k + w_k, \\ y_k &= Cx_k + v_k, \end{aligned}$$

and estimates the full state x_k, not just a filtered version of y_k.

Frequency-domain vs model-based design. Classical filters are

- designed in the frequency domain, and
- require assumptions about noise frequency content,

where as Kalman filter is

- designed using a state-space model,
- uses statistical assumptions (covariances Q, R), and
- takes advantage of the system dynamics to infer hidden states.

No notion of system dynamics. A low-pass filter cannot infer velocity from position. The Kalman filter can reconstruct unmeasured states by exploiting dynamics

$$\dot{p} = v,$$

allowing velocity estimation from noisy position measurements.

Adaptive weighting. Classical filters apply fixed filtering. The Kalman filter automatically adjusts the weighting between prediction and measurement through the Kalman gain

$$K_k = P_k C^T (C P_k C^T + R_k)^{-1}.$$

A Kalman filter is not a better low-pass filter. It is a fundamentally different object. It is a *model-based estimator* that uses system dynamics to reconstruct the state from noisy measurements.

4.4 Discrete-Time System Model

We consider the discrete-time linear system

$$x_{k+1} = Ax_k + Bu_k + w_k, \tag{54}$$
$$y_k = Cx_k + v_k, \tag{55}$$

where

- $x_k \in \mathbb{R}^n$ is the state at time step k,
- $u_k \in \mathbb{R}^m$ is the input,
- $y_k \in \mathbb{R}^p$ is the measurement,
- w_k is process noise,
- v_k is measurement noise.

4.5 Discretization of Continuous-Time Models

In many applications, the system is originally modeled in continuous time. However, since measurements and control inputs are updated at discrete time instants, we convert the model to discrete time.

Let the sampling time be Δt. Then, defining

$$x_k \triangleq x(k\Delta t), \quad u_k \triangleq u(k\Delta t),$$

the discrete-time model can be written as

$$x_{k+1} = A_d x_k + B_d u_k. \tag{56}$$

4.5.1 Exact Discretization

The exact discrete-time matrices are given by

$$A_d = e^{A\Delta t}, \quad B_d = \int_0^{\Delta t} e^{A\tau} B \, d\tau. \tag{57}$$

This discretization is exact under the assumption that the input $u(t)$ is constant over each sampling interval $[k\Delta t, (k+1)\Delta t)$ (zero-order hold). The matrices A_d and B_d can be computed numerically using `c2d`. For example,

```
sysd = c2d(ss(A,B,C,D), Ts, 'zoh');
```

computes the exact discrete-time model under a zero-order hold assumption.

4.5.2 Forward Euler Approximation

For small sampling times Δt, a simple approximation is

$$x_{k+1} \approx x_k + \Delta t(Ax_k + Bu_k), \tag{58}$$

which gives

$$A_d \approx I + \Delta t A, \tag{59}$$

$$B_d \approx \Delta t B. \tag{60}$$

Note that a smaller Δt implies better approximation, and is commonly used in simulations and simple implementations.

4.6 Noise Characterization

In practice, neither the system dynamics nor the measurements are exact. In the model (54), (55), w_k and v_k represent uncertainty in the model and measurements, respectively.

The **process noise** w_k captures uncertainty in the system dynamics. This may arise due to unmodeled dynamics (e.g., neglected nonlinearities), external disturbances (e.g., wind, friction, vibrations), parameter uncertainty (e.g., mass, inertia), and discretization errors. Thus, even if the model is known, the true state evolution deviates from the predicted model.

The **measurement noise** v_k represents errors in sensing. This may arise due to sensor noise (electrical or thermal), quantization errors, bias and calibration errors, and limited sensor resolution. Thus, the measured output y_k is only an approximation of the true output.

4.6.1 Random Variables

A random variable $x \in \mathbb{R}^n$ is a vector whose value is uncertain. The **mean** or **expected value** of x is defined as

$$\mathbb{E}[x] = \int_{\mathbb{R}^n} x\, p(x)\, dx, \tag{61}$$

where $p(x)$ is the probability density function of x. The **covariance** of x is defined as

$$\text{cov}(x) = \mathbb{E}\big[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^T\big]. \tag{62}$$

The mean represents the average value of the random variable, and the covariance quantifies the uncertainty and correlations between components.

4.6.2 Statistical Model

We model the noise as random variables

$$w_k \sim \mathcal{N}(0, Q), \tag{63}$$
$$v_k \sim \mathcal{N}(0, R), \tag{64}$$

and assume that w_k and v_k are independent, noise is zero-mean (no systematic bias), and noise is uncorrelated across time.

The matrices Q and R quantify the level of uncertainty in the model and measurements, respectively. The process noise covariance $Q \succeq 0$ reflects how much we trust the system dynamics: larger values of Q indicate greater uncertainty in the model and lead the estimator to rely more heavily on measurements, while smaller values imply higher confidence in the model predictions. Similarly, the measurement noise covariance $R \succ 0$ captures the quality of the sensor data: larger values of R correspond to noisier measurements and reduce their influence on the estimate, whereas smaller values indicate more reliable measurements. Thus, the relative magnitudes of Q and R determine how the estimator balances model-based prediction and measurement correction.

4.7 Estimator Structure

The Kalman filter consists of two steps

- Prediction (model-based),
- Update (measurement correction).

4.7.1 Prediction Step

At time step $k+1$, we define two estimates of the state x_{k+1},

- the **prior estimate** $\hat{x}_{k+1|k}$, which is obtained by prediction from step k, and
- the **posterior estimate** $\hat{x}_{k+1|k+1}$, which is obtained by incorporating the measurement y_{k+1},

Similarly, we define the associated error covariance matrices

- the **prior covariance** $P_{k+1|k}$, which quantifies uncertainty in the predicted state, and

- the **posterior covariance** $P_{k+1|k+1}$, which quantifies uncertainty after incorporating the measurement.

Given the posterior estimate $\hat{x}_{k|k}$ at step k, we compute the prior estimate at step $k+1$ as

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k} + Bu_k. \tag{65}$$

The corresponding covariance is propagated as

$$P_{k+1|k} = AP_{k|k}A^T + Q. \tag{66}$$

The covariance propagation reflects how uncertainty evolves. The term APA^T propagates existing uncertainty, while Q adds new uncertainty due to process noise.

4.7.2 Update Step

When a new measurement y_{k+1} becomes available, the posterior estimate at step $k+1$ is computed as

$$\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k} + K_{k+1}r_{k+1}, \tag{67}$$

where the **Kalman Gain** is

$$K_{k+1} = P_{k+1|k}C^T(CP_{k+1|k}C^T + R)^{-1}, \tag{68}$$

and the **Innovation** is

$$r_{k+1} = y_{k+1} - C\hat{x}_{k+1|k}. \tag{69}$$

The corresponding covariance is propagated as

$$P_{k+1|k+1} = (I - K_{k+1}C)P_{k+1|k}. \tag{70}$$

The Kalman gain determines how the estimator balances model prediction and measurement correction. If the predicted uncertainty is large (i.e., P is large), then the gain K becomes large, placing greater weight on the measurements. Conversely, if the measurement noise is large (i.e., R is large), then the gain K becomes small, reducing the influence of the measurements and placing more trust in the model. Thus, the Kalman gain automatically adjusts the relative confidence in prediction versus sensing.

4.8 Optimality of the Kalman Filter

Note that the Kalman filter has the same structure as an observer

$$\hat{x}_{k+1} = A\hat{x}_k + Bu_k + K_k(y_k - C\hat{x}_k), \tag{71}$$

but the gain K_k is computed optimally based on the noise covariances Q and R. The Kalman filter is, in fact, the optimal linear estimator under the assumed noise model [2]. In particlar, the Kalman filter computes the estimate $\hat{x}_{k|k}$ that minimizes the mean-square estimation error

$$\hat{x}_{k|k} = \arg\min_{\hat{x}} \ \mathbb{E}\big[\|x_k - \hat{x}\|^2\big]. \tag{72}$$

4.9 Estimation of Measured States

A common misconception is that the Kalman filter is only useful for estimating states that are not directly measured. In reality, the Kalman filter can improve even directly measured states because the measurements are noisy.

Consider the scalar system

$$x_{k+1} = x_k + w_k, \tag{73}$$
$$y_k = x_k + v_k, \tag{74}$$

where $w_k \sim \mathcal{N}(0, Q)$, and $v_k \sim \mathcal{N}(0, R)$. Note that the state is directly measured since $C = 1$. However, the measurement y_k is not equal to x_k because of measurement noise.

The Kalman filter prediction step is

$$\hat{x}_{k+1|k} = \hat{x}_{k|k}, \tag{75}$$
$$P_{k+1|k} = P_{k|k} + Q. \tag{76}$$

When the measurement y_{k+1} becomes available, the update step is

$$K_{k+1} = \frac{P_{k+1|k}}{P_{k+1|k} + R}, \tag{77}$$
$$\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k} + K_{k+1}\left(y_{k+1} - \hat{x}_{k+1|k}\right), \tag{78}$$
$$P_{k+1|k+1} = (1 - K_{k+1})P_{k+1|k}. \tag{79}$$

Substituting the Kalman gain into the covariance update gives

$$P_{k+1|k+1} = \left(1 - \frac{P_{k+1|k}}{P_{k+1|k} + R}\right) P_{k+1|k} \tag{80}$$
$$= \frac{R}{P_{k+1|k} + R} P_{k+1|k}. \tag{81}$$

Since

$$0 < \frac{R}{P_{k+1|k} + R} < 1, \tag{82}$$

we obtain

$$P_{k+1|k+1} < P_{k+1|k}. \tag{83}$$

Thus, even though x_k is directly measured, incorporating the measurement reduces the state uncertainty.

Remark 1. The estimate is not simply the measurement. It is the weighted average

$$\hat{x}_{k+1|k+1} = (1 - K_{k+1})\hat{x}_{k+1|k} + K_{k+1} y_{k+1}. \tag{84}$$

Even when the full state is measured, the measurement is noisy. The Kalman filter improves the estimate by combining the noisy measurement with the model-based prediction and reducing the estimation uncertainty.

References

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[2] A. Goel and D. S. Bernstein, "On the Accuracy of the One-step UKF and the Two-step UKF," in *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 5375–5380, IEEE, 2022.